

1. From the top A of a vertical wall AB of height 30 m, the angles of depression of the top P and bottom Q of a vertical tower PQ are 15° and 60° respectively, B and Q are on the same horizontal level. If C is a point on AB such that $CB = PQ$, then the area (in m^2) of the quadrilateral $BCPQ$ is equal to

[2023 (06 Apr Shift 1)]

(1) $300(\sqrt{3} - 1)$

(2) $300(\sqrt{3} + 1)$

(3) $600(\sqrt{3} - 1)$

(4) $200(\sqrt{3} - 1)$

2. The angle of elevation of the top P of a tower from the feet of one person standing due south of the tower is 45° and from the feet of another person standing due west of the tower is 30° . If the height of the tower is 5 meters, then the distance (in meters) between the two persons is equal to

[2023 (11 Apr Shift 2)]

(1) $\frac{5}{2}\sqrt{5}$

(2) 10

(3) 5

(4) $5\sqrt{5}$

ANSWER KEYS

1. (3)

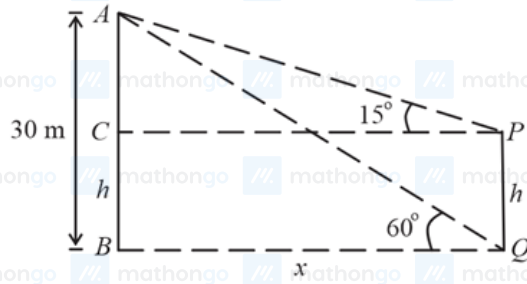
2. (2)

1. (3)

Given that from the top A of a vertical wall AB of height 30 m, the angles of depression of the top P and bottom Q of a vertical tower PQ are 15° and 60° respectively,

B and Q are on the same horizontal level and C is a point on AB such that $CB = PQ$.

The required diagram will be



From the diagram,

$$\Rightarrow \tan 60^\circ = \frac{AB}{BQ}$$

$$\Rightarrow \frac{30}{x} = \sqrt{3};$$

$$\Rightarrow x = 10\sqrt{3} \text{ m}$$

$$\text{Also, } \tan 15^\circ = \frac{AC}{CP}$$

$$\Rightarrow \frac{30-h}{x} = 2 - \sqrt{3}$$

$$\Rightarrow 30 - h = 10\sqrt{3}(2 - \sqrt{3})$$

$$\Rightarrow 30 - h = 20\sqrt{3} - 30$$

$$\Rightarrow h = 60 - 20\sqrt{3}$$

Now area of the quadrilateral $BCPQ$ is $BC \times BQ$

$$\therefore \text{area} = hx$$

$$= (60 - 20\sqrt{3})10\sqrt{3}$$

$$= 200\sqrt{3}(3 - \sqrt{3})$$

$$= 600(\sqrt{3} - 1) \text{ m}^2$$

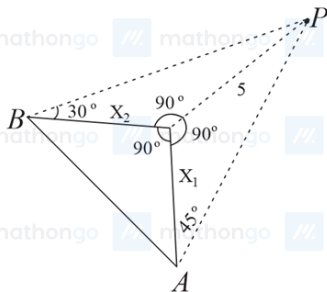
Hence, this is the correct option.

2. (2)

Given that the elevation to the top of the tower from the feet of one person due south is 45°

Similarly the angle of elevation to the top of tower from the feet of another person to the west of tower is 30°

The given information can be represented in the form of a diagram as below:



We need to find the distance AB .

Let the distance of A, B from the tower be x_1, x_2 .

$$\Rightarrow \tan 45^\circ = \frac{5}{x_1}, \tan 30^\circ = \frac{5}{x_2}$$

$$\Rightarrow x_1 = 5, x_2 = 5\sqrt{3}.$$

$$\Rightarrow AB = \sqrt{x_1^2 + x_2^2} = \sqrt{(5)^2 + (5\sqrt{3})^2} = 10 \text{ m}$$

Hence this is the correct option.